

2. (10 points) We have 2 boxes, each containing 3 balls. Box number 1 contains one black and two white balls; box number 2 contains two black and one white ball.

Our friend chooses one of the boxes at random, probability of choosing box number 1 is p . Then he takes one ball from a chosen box (each of three balls can be taken chosen equally likely), and it turns out to be white.

We are going to find MAP estimate for the parameter $\theta = 1$ or 2 to conclude which box was chosen (θ is the number of the box). For what values of p we can conclude that box number 1 was chosen? For what values of p we can conclude that box number 2 was chosen?

Solution

Prior density (PMF) of choosing each box is $P(\text{Box} = 1) = p$, $P(\text{Box} = 2) = 1 - p$ (Box is the name of our parameter, it can take two values, 1 or 2, given the value of the Box we know conditional probabilities to draw a white and a black ball). Bayes theorem gives:

$$\begin{aligned} \mathbb{P}(\text{Box} = 1 | \text{Ball} = \text{white}) &= \\ \frac{\mathbb{P}(\text{Ball} = \text{white} | \text{Box} = 1) \cdot \mathbb{P}(\text{Box} = 1)}{\mathbb{P}(\text{Ball} = \text{white} | \text{Box} = 2) \cdot \mathbb{P}(\text{Box} = 2) + \mathbb{P}(\text{Ball} = \text{white} | \text{Box} = 1) \cdot \mathbb{P}(\text{Box} = 1)} &= \\ \frac{2p/3}{2p/3 + (1-p)/3} &= \frac{2p}{1+p}. \end{aligned}$$

Then,

$$\mathbb{P}(\text{Box} = 2 | \text{Ball} = \text{white}) = 1 - \frac{2p}{1+p} = \frac{1-p}{1+p}.$$

MAP estimate for the parameter Box is the point of maximum of the posterior density, which takes only 2 values: $\frac{2p}{1+p}$ and $\frac{1-p}{1+p}$. So, MAP estimate gives $\text{Box} = 1$ if

$$\frac{2p}{1+p} > \frac{1-p}{1+p},$$

that is, $2p > 1 - p$, $p > 1/3$. If $p < 1/3$, we would conclude (based on the MAP estimate) that box 2 was chosen. When $p = 1/3$ we cannot decide since posterior density is constant (and have several points of argmax).